

The Kinematics and Dynamics of Complex Crater Collapse

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1 Introduction

Large impact craters on rocky planetary bodies possess central peaks or a ring of peaks within the crater rim (Roddy, 1977). Impact structures with these morphologies are known as complex craters. In addition to central peaks or peak-rings, complex craters have substantially larger aspect ratios than the smaller, bowl-shaped, and so-called simple craters (Figure 1.1; Melosh, 1989). The transition from simple to complex craters occurs at a critical size threshold, which is inversely proportional to the gravitational acceleration on the cratered body (Pike, 1977). At, and beyond that critical size threshold, craters begin to display centrally uplifted peaks and/or rings and possess wide, terraced rim areas.

The dependence of the critical size threshold on gravity for the onset of complex crater formation indicates that the collapse of complex craters occurs due to weight overcoming the strength of rocks in the transient cavity. Static and dynamic models have indicated that a remarkable reduction of cohesion and internal friction is required for crater collapse (McKinnon, 1978; Melosh, 1977; O’Keefe and Ahrens, 1999; Wünnemann and Ivanov, 2003); the effective strength of impacted rocks must be less than ~ 3 MPa and there must be little or no internal friction. These properties are inconsistent with laboratory measurements of the properties of rocks and even those of damaged rocks. Therefore, there must be an additional mechanism or mechanisms that transiently and significantly reduce rock strength such that complex crater collapse is facilitated. Several mechanisms have been suggested to contribute to this significant and transient weakening; fracturing and rock damage (e.g. Collins et al., 2004; Melosh et al., 1992), bulk thermal softening (O’Keefe and Ahrens, 1999), fault weakening (Senft and Stewart, 2009) by shear-induced thermal softening (Spray, 1997) and other processes, and acoustic fluidisation (Melosh, 1979).

Resolving which, if any, of these mechanisms act during complex crater formation requires observational and/or experimental constraints. These constraints include not only direct measurements and observations of the potential products of those weakening mechanisms, but also indirect, geological, constraints that describe the bulk behaviour of rocks during crater collapse, i.e. kinematics.

with the shock and rarefaction waves. During cratering, it is this remnant velocity that causes the growth of a transient cavity. The conditions of stress described here are diagrammatically shown in Figure 2.3. Consideration of the orientation of stresses during cratering, where the shock wave is roughly hemispherical, is considerably more complex; this is because rarefaction does not propagate parallel to the shock front and because of the effect of the expansion of the shock front at increasing radial distances.

The contact and compression stage ends after the projectile has unloaded from high pressure. The principal result of this stage is the transfer of most of the projectile’s kinetic energy to the target. The contact and compression stage is typically over within a second, even for the largest of planetary impacts (Melosh, 1989).

2.1.2 Excavation

The excavation stage begins immediately after the contact and compression stage ends. During this stage, the shock wave propagates into the target, attenuating as it is transmitted due to the increased area of the shock front and eventually, due to the rarefaction wave catching up with the shock front. The combined effect of the shock and rarefaction waves is to accelerate the material such that a crater is opened by the excavation and ejection of material, as well as the displacement of material deeper into the target. The excavation stage typically lasts many times the length of the contact and compression stage, requiring seconds or minutes to reach completion, depending on the crater size (Melosh, 1989).

2.1.3 Modification

The modification stage begins after the crater has been fully excavated. The result of the excavation stage is a bowl-shaped “transient crater” which then, usually, collapses under gravity. The extent to which a crater undergoes modification is strongly dependent on the size of the transient crater. In small craters, debris flows slide down the interior walls of the transient cavity. In large craters, rim areas are terraced due to the slumping of large blocks at the rim while rocks are uplifted near to the crater centre, resulting in central peaks and/or peak-rings. These large craters are the focus of this thesis and will be described in greater detail in Section 2.2. This stage of cratering may last more than an order of magnitude longer than the excavation stage (Melosh, 1989).

2.2 Complex Crater Collapse

Large impact craters have complex morphologies, containing central peaks or peak-rings and wide terrace zones at their rim. By contrast, small impact craters have bowl-shaped, and so-called “simple” morphologies (Figure 1.1). Complex crater morphologies form during the collapse of a transient, bowl-shaped cavity. The diameter at which craters transition from simple to complex morphologies occurs over a narrow range, and is specific to each planetary body. On the Moon, complex craters can be observed at about 15 km diameter. On Mercury and Mars the transition occurs at about 7 km diameter, and on Earth, the transition occurs at about 2–4 km diameter, depending on the target material. It can therefore be seen that the simple-to-complex transition diameter scales inversely with the acceleration due to gravity on the given planetary body (Figure 2.4; Pike, 1980, 1988).

The dependence of the simple-to-complex transition on gravitational acceleration suggests that craters collapse due to weight, the force generated by the effect of gravity on mass, overcoming the strength of rocks i.e. the ability of the transient cavity to maintain its shape under stress. Static analysis of the stability of cavities has shown that for substantial rim collapse or floor uplift to occur, the actual effective strength must be less than 3 MPa, and there must be little or no internal friction (McKinnon, 1978; Melosh, 1977). Additionally dynamic models of crater formation require the same low yield strength and internal friction (O’Keefe and Ahrens, 1999; Wünnemann and Ivanov, 2003). These properties are inconsistent with laboratory measurements of the strength of rock (Kenkmann et al., 2012), thus some process must occur that reduces the effective strength of rocks during cratering. A variety of mechanisms could potentially cause the weakening of rocks within craters: damage and fracturing (see Kenkmann et al., 2012), bulk thermal softening (O’Keefe and Ahrens, 1999), fault lubrication by shear-induced melting (Spray, 1997) or other processes (Senft and Stewart, 2009), Acoustic Fluidisation (Melosh and Ivanov, 1999; Melosh, 1979), or some combination of these effects.

Damage and rock fracturing cannot be solely responsible for the required weakening. Figure 2.5 shows the damaged and intact strength profiles of a typical rock (see Section 3.2.2.2). It can be seen that while damage can reduce the shear strength of a rock to significantly less than several MPa, damage is only capable of causing sufficient weakening at normal stresses (i.e. pressures) less than approximately 10 MPa, equivalent to only 300 m of lithostatic overburden. The vast majority of rocks at failure during complex crater collapse are under normal stresses considerably greater than 10 MPa. In other words, the coefficient of internal friction of damaged rocks is too high, even for damaged and fractured rocks.

It has been proposed that the deposition of energy in the target due to the passage of the shock wave may cause the thermal softening of rocks (O’Keefe and

7 Introduction to the Chicxulub Impact Structure

The Chicxulub Crater, located in the Yucatán peninsula of Mexico, is a ~ 180 km diameter impact structure. The identification of the Chicxulub structure (Hildebrand et al., 1991; Penfield and Camargo, 1981) was made on the basis of large-scale geophysical signatures: a large negative Bouguer gravity anomaly and magnetic anomalies. The crater is buried, and the only surface expression of its presence is a 180 km diameter ring of cenotes, sinkholes in the Cenozoic limestones that overlie the crater.

Following the discovery of the crater, several attempts were made to constrain the size and structure of the Chicxulub based on forward modelling of potential field data (Espindola et al., 1995; Hildebrand et al., 1995; Hildebrand et al., 1991; Sharpton et al., 1993, 1996). The variation between these models, in terms of structure and the density contrasts between lithologies are large as a consequence of the non-uniqueness of gravity and magnetic modelling, for example, the estimated density contrasts between fractured basement rocks and intact crustal rocks vary between 0 kg m^{-3} and 490 kg m^{-3} , assuming a minimum density of deep crustal rocks to be 2600 kg m^{-3} , this corresponds to porosities from 0%, up to 19%.

Seismic reflection and refraction surveys were carried out at the site in order to resolve the crater and sub-crater structure (Gulick et al., 2008; Morgan et al., 1997). The results of those surveys (Gulick et al., 2013 and refs therein) indicated that the Chicxulub crater possesses:

1. A topographic (inner) rim, ~ 150 km in diameter, hereafter termed the “crater rim”;
2. Non-continuous ring-faults, external to the crater rim; one, the “outer” ring ~ 200 km in diameter, and a second, “exterior” ring ~ 250 km in diameter;
3. An 80–90 km diameter peak-ring which rises up to 400 m above the crater floor;
4. A strong reflector beneath the peak-ring which dips towards the crater centre;
5. A low-velocity zone, $\sim 4 \text{ km s}^{-1}$, associated with the peak-ring, which continues downwards and inwards, towards the crater centre, parallel to the dipping reflector;

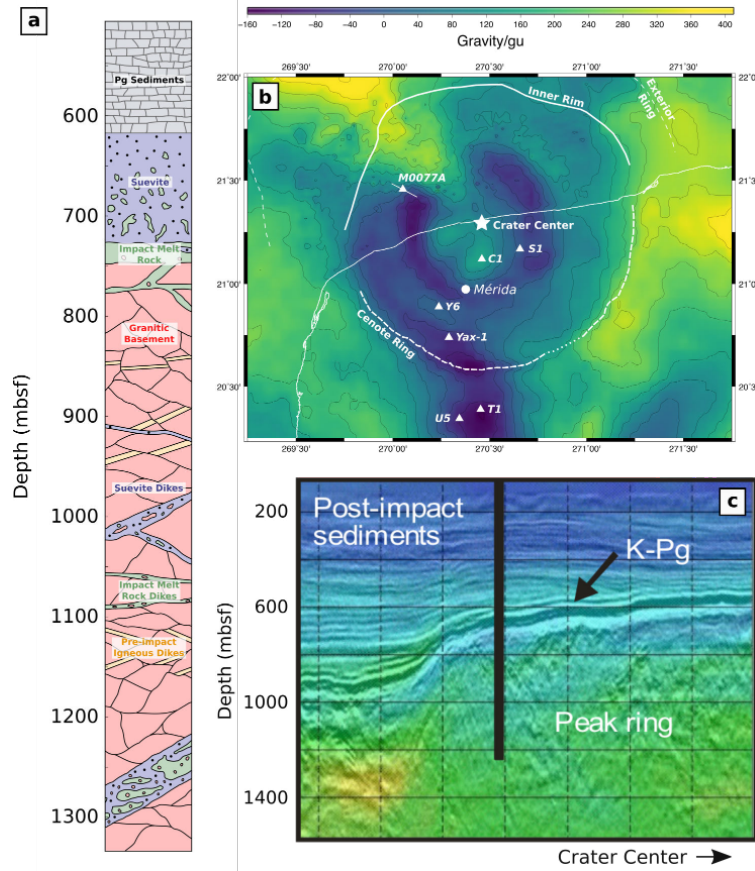


Figure 7.2: a) Schematic stratigraphic log of Hole M0077A. The location of M0077A is indicated on subfigure b) a Bouguer gravity anomaly map of the crater, and c) a radial seismic profile of the peak-ring through the site location. Modified from Gulick et al. (2013) and Morgan et al. (2016).

tion recovered core between 505.42 and 1363.05 mbsf (metres below sea floor) including: from 505.42 - 617.82 mbsf, a sequence of carbonate-rich Palaeogene sedimentary rocks, from 617.82 - 750.25 mbsf, a sequence of allochthonous impactites, predominantly fining-upward suevitic breccias (a polymict breccia containing clasts of impact melt rock/glass) with impact melt rock at its base, and from 750.25 - 1363.05 mbsf, a sequence of uplifted granitic target rocks, with occasional dikes of pre-impact igneous rocks (porphyritic and non-porphyritic dolerites, porphyritic dacite, and a homogeneous fine-grained felsite) and impact-related breccias and melt rocks. A schematic lithological column and location map is shown in Figure 7.2.

Initial analysis of the recovered core provided several constraints on the provenance of peak-ring material and the deformation that those rocks must have experienced during cratering. The occurrence of crystalline rocks at depths of only 750 mbsf

suggests that the crystalline rocks have been uplifted by at least 2.25 km, such that rocks derived from beneath the 3 km-thick Mesozoic sedimentary sequence reach their final location in the peak-ring. Additionally, the presence of abundant quartz PDFs and no diaplectic quartz glasses within the granitic target rocks indicate that the recovered para-autochthon, the shocked target rocks, experienced shock pressures from 10–35 GPa (Morgan et al., 2016). Additionally physical property measurements (Christeson et al., 2018; Morgan et al., 2017) has indicated that the shocked target rocks within the Chicxulub peak-ring has porosities of approximately 10%, a likely contributor to the gravity minimum associated with the peak-ring. Finally, constraints have also been applied to the structural history of the peak-ring rocks by Riller et al. (in press) (see Morgan et al., 2017). Initially, target rocks were pervasive fractured followed, sequentially by the formation of cataclasite zones, ultra-cataclasite zones, shear faults, the emplacement of impact breccias and impact melts into dilatant fractures, and finally, the formation of ductile band structures.

$${}^t\boldsymbol{\sigma} = \begin{bmatrix} P & 0 & 0 \\ 0 & P & 0 \\ 0 & 0 & P \end{bmatrix} + \begin{bmatrix} \sigma_{rr} - P & \sigma_{rz} & 0 \\ \sigma_{rz} & \sigma_{zz} - P & 0 \\ 0 & 0 & \sigma_{\theta\theta} - P \end{bmatrix} = {}^t\mathbf{P} + {}^t\mathbf{s} , \quad (8.13)$$

where ${}^t\mathbf{P}$ is the hydrostatic stress tensor, which contains diagonal elements that each equal to the pressure, $P = \frac{tr({}^t\boldsymbol{\sigma})}{3}$. ${}^t\mathbf{s}$ is the deviatoric stress tensor. The deviatoric stress tensor possesses eigenvectors that are parallel to those of the stress tensor, and eigenvalues, s_1 , s_2 , and s_3 , that are related to the principal stresses by the relationship:

$$s_{1,2,3} = \sigma_{1,2,3} - P , \quad (8.14)$$

s_1 , s_2 , and s_3 are known as the deviatoric principal stresses.

Furthermore, there are three invariants of the deviatoric stress tensor, denoted J_1 , J_2 , and J_3 . These invariants can be determined following equations similar to Equations (8.8), (8.9), and (8.10), respectively. J_2 is of particular physical significance as it is a measure of the deviatoric part of the stress tensor i.e. it is a measure of the shear stress. Nevertheless, $\sqrt{J_2}$ is commonly used because it has units of Pa .

8.3 Results

The above analysis was used to examine the stress-strain paths of material within a simulation of the Chicxulub impact event. Of the para-autochthonous rocks within a large impact structure, rocks within the peak-ring experience the most extreme deformation during modification. For this reason, and due to the recent IODP-ICDP drilling of the Chicxulub peak-ring, the peak-ring materials will be focussed upon specifically. Here, peak-ring material is defined based on a number of criteria:

1. Peak-ring material in the Chicxulub Crater occurs no further than 45 km from the crater centre and no less than 30 km;
2. Peak-ring material must be solid at the end of the simulation. Melted material is distinguished by peak shock pressures > 60 GPa.
3. Peak-ring material must derive from within the central uplift, this is defined at the time of maximum central uplift (~ 200 s for a Chicxulub-sized simulation on Earth). The central uplift is defined as any material within the trough that surrounds the uplift at this time, and within 10 km of the free surface.

The location of the material that satisfies these three criteria, the peak-ring rocks, is shown on Figure 8.2 throughout the simulation.

The peak-ring originates from 5-12 km depth, and 10-30 km radial distance (Figure 8.2a). After shock, decompression, and transient cavity growth, the material moves outwards, into the transient cavity wall (Figure 8.2b). This material subsequently collapses inwards (Figure 8.2c) and becomes entrained into the flank of the central uplift (Figure 8.2d). At this time, the central uplift is over-heightened, and collapses radially outwards (Figure 8.2e), emplacing the peak-ring on top of the collapsed transient cavity rim (Figure 8.2f). The results presented here will use a single Lagrangian pseudo-cell from within the centre of the peak-ring material, and is thus representative of the stress-strain history of the entire peak-ring. This pseudo-cell originates from 20.2 km radial distance and 8.95 km depth, and ends at 35 km radial distance and 2.5 km depth. The sequence of deformation of this cell will be described chronologically in the following sections.

8.3.1 Shock and Decompression

The first stage of impact related deformation to affect the peak-ring rocks is shock and decompression. The strain pathway of the selected pseudo-cell during this stage is shown in Figure 8.3. Deformation is characterised by an initial, rapid horizontal compression of the material as it is shocked, this results in a sharp reduction in the volume of the material (negative volumetric strains), and a sharp increase in shear strain. During these few milliseconds, strain rate reaches its maximum throughout the entire simulation, 0.17 s^{-1} , n.b. this value may be a slight underestimate due to:

1. Artificial viscosity (see 3.2.1.5);
2. Spatial and temporal resolution;
3. The time interval between saved timesteps, which during this time of the simulation is 0.1 s.

As decompression occurs, the strain rate rapidly drops, volumetric strain increases and becomes positive (due to pressure release). Throughout the entire stage of deformation, there is a negligible amount of rotation of the material from its initial orientation.

During shock and decompression, pressures and shear stresses rapidly increase. For the selected pseudo-cell, the arrival of the stress wave occurs at 2.10 s (Figure 8.4).

Most of the peak-ring material never experiences shock pressures greater than $\sim 40 \text{ GPa}$, therefore the peak-ring material experiences shock metamorphism from a plastic wave with an elastic precursor. The stress wave exceeds the Hugoniot Elastic Limit (HEL; 3.56 GPa for the granite strength properties used here) at 2.43 s where